

Investigation of Near-Accident Car-Driving Scenarios using Deep Imitation Learning + RL

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Inference & Performance Report

Project Title: Investigation of Near-Accident Scenarios using a Hybrid Deep Learning Approach

Model ID: rl_model.zip (PPO, fine-tuned)

Date of Evaluation: October 13, 2025

Environment: highway-env

1. Executive Summary

This report details the performance of an autonomous driving agent trained via a hybrid Imitation Learning (IL) and Reinforcement Learning (RL) methodology. The agent's objective was to navigate a dynamic highway simulation, prioritizing collision avoidance and efficiency. Evaluation confirms the success of the hybrid strategy: the initial IL model achieved a high but misleading accuracy of 88.10% due to data imbalance, while the final RL-tuned agent demonstrated superior performance, achieving a mean episode reward of 29.87. This represents a 155% improvement over its initial reward score, validating that RL fine-tuning is crucial for developing robust driving policies.

2. Model & Environment Details

- *Algorithm: Proximal Policy Optimization (PPO).*
- *Architecture: A deep Convolutional Neural Network (CNN) feature extractor followed by a multi-layer perceptron policy network.*
- *Training Regimen: The model was first pre-trained via Imitation Learning on a human-generated dataset, then fine-tuned with Reinforcement Learning for 20,000 timesteps.*
- *Evaluation Environment: highway-v0 configured with 50 vehicles and a high collision penalty to simulate near-accident conditions.*

3. Quantitative Performance Metrics

Final Reinforcement Learning Model Performance

This model was evaluated on its ability to maximize cumulative reward over 20 test episodes. A higher reward indicates safer and more efficient driving.

Metric	Value	Interpretation
Mean Reward per Episode	29.87	Primary performance score. A strong result demonstrating effective driving.
Std. Dev. of Reward	9.48	High variability. Indicates the agent performs very well in some runs but struggles in others, pointing to specific unmastered scenarios.
Training Improvement	11.7 → 27.0	The agent's average score more than doubled during RL training, providing clear evidence of successful learning.

Baseline Imitation Learning Model Performance

This model was evaluated on its ability to correctly predict the expert's action on a test set of 269 samples.

Metric	Score	Interpretation
Overall Accuracy	88.10%	High but misleading. Achieved primarily by predicting the most common action ("IDLE").
F1-Score (Weighted Avg)	0.86	A more reliable score, confirming a strong but imperfect baseline.

4. Qualitative Analysis & Key Findings

- 1. RL Outperforms IL Significantly:** The final agent's mean reward of 29.87 is a clear improvement over the IL baseline, which was hesitant and failed to learn critical actions like accelerating or turning left. RL successfully taught the agent to be more decisive and proactive.
- 2. Inconsistency Reveals Edge Cases:** The high standard deviation (9.48) is a key finding. It reveals that while the agent mastered common driving patterns (achieving high scores of ~36), it is not yet fully robust. Certain dense traffic or sudden "near-miss" situations likely caused the low-scoring runs (e.g., 4.40, 14.13), highlighting specific weaknesses.
- 3. Data Imbalance Limits Imitation Learning:** The IL model's evaluation showed a heavy bias toward the "IDLE" action (86% of the dataset). This proves that pure imitation is insufficient for learning rare but critical maneuvers, making the RL fine-tuning phase essential.

5. Conclusion & Recommendations

The hybrid IL+RL approach was highly effective. Imitation Learning provided a solid foundation, and Reinforcement Learning successfully enhanced the agent's capabilities, resulting in a more competent driver.

- *Conclusion: The project successfully demonstrates a methodology for training agents to handle complex driving scenarios, with the final model showing clear intelligence and adaptability.*
- *Next Steps:*
 - *Extended Training: Continue RL training to further improve the mean reward and, crucially, reduce the standard deviation.*
 - *Reward Shaping: Adjust the reward function (e.g., add a penalty for jerky movements) to refine the agent's driving style.*
 - *Diverse Data Collection: Collect a more balanced dataset for the IL phase to improve the initial baseline.*